CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

# Chapter 11: Areas Related to Circles

Every formula and result of the chapter on one sheet

| | | |
|---|---|---|
| 1 | Circumference of a circle | $2\pi r$ |
| 2 | Area of a circle | $\pi r^2$ |
| 3 | Area of a sector of angle $\theta$ | $\frac{\theta}{360} \times \pi r^2$ |
| 4 | Length of an arc of angle $\theta$ | $\frac{\theta}{360} \times 2\pi r$ |
| 5 | Area of a sector from its arc length $l$ | $\frac{1}{2} l\, r$ |
| 6 | Perimeter of a sector | $\frac{\theta}{360} \times 2\pi r + 2r$ |
| 7 | Area of the major sector | $\pi r^2 -$ area of the minor sector |
| 8 | Area of a minor segment | area of sector $-$ area of triangle |
| 9 | Area of the major segment | $\pi r^2 -$ area of the minor segment |
| 10 | Triangle in a sector, $\theta = 60^\circ$ | $\frac{\sqrt{3}}{4} r^2$ |
| 11 | Triangle in a sector, $\theta = 90^\circ$ | $\frac{1}{2} r^2$ |
| 12 | Triangle in a sector, $\theta = 120^\circ$ | $\frac{\sqrt{3}}{4} r^2$ |
| 13 | Area of a ring | $\pi(R^2 - r^2)$ |
| 14 | Angle turned by the minute hand | $6^\circ$ per minute |
| 15 | Angle turned by the hour hand | $30^\circ$ per hour, $0.5^\circ$ per minute |